

Nmap Complete Guide

Beginner se Advanced tak - Hindi mein samjhaya gaya

Yeh guide sirf educational aur learning purpose ke liye hai. Nmap ka use hamesha apne khud ke network/devices par ya jahan explicit legal permission ho, wahi karein. Bina permission kisi doosre ke network ko scan karna illegal ho sakta hai.

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1. Nmap Kya Hai Aur Kyun Use Hota Hai

Nmap (Network Mapper) ek free aur open-source tool hai jo network scanning ke liye use hota hai. Iska main kaam hai kisi bhi device (computer, router, phone, server) ke baare mein jaankari nikalna - jaise woh device online hai ya nahi, usme kaunse ports khule hain, kaunsi services chal rahi hain, aur kabhi kabhi operating system kaunsa hai.

Socho ek building hai (IP address), aur us building mein bahut saare darwaze hain (ports). Nmap ek tarah se har darwaze ko knock karke check karta hai ki woh khula hai ya band, aur khule darwaze ke peeche kaun (konsi service) baitha hai.

Nmap kis kis field mein use hota hai

- Network administrators apne network ko monitor karne ke liye
- Security professionals (pentesters) vulnerabilities dhundne ke liye
- Students cybersecurity seekhne ke liye apne practice lab mein
- System admins yeh check karne ke liye ki koi unwanted service toh nahi chal rahi

2. Nmap Install Karna

Kali Linux mein Nmap already pre-installed hota hai. Phir bhi agar check karna ho ya install/update karna ho:

```
nmap --version
```

Yeh command batata hai ki Nmap already installed hai ya nahi, aur uska version kya hai.

```
sudo apt update  
sudo apt install nmap -y
```

Pehli line se system ke saare software packages ki latest list update hoti hai. Doosri line se Nmap install ho jata hai (-y ka matlab hai har confirmation ko automatically 'yes' bol dena).

3. Basic Scanning Commands

3.1 Simple scan

```
nmap <target>
```

Yahan <target> ki jagah koi IP address ya domain naam daala jata hai. Yeh command default tareeke se sabse common 1000 ports check karta hai target par.

3.2 Poora subnet/network scan karna

```
nmap <network>/24
```

Jaise example ke liye agar range likhi jaye /24, toh yeh us poore local network ke saare devices (256 tak IP addresses) ko scan kar deta hai. Yeh sirf apne khud ke network par hi try karna chahiye.

3.3 Multiple targets ek saath scan karna

```
nmap 10.0.0.1 10.0.0.2 10.0.0.3
```

Space se separate karke multiple IP addresses ek hi command mein scan kar sakte hain.

```
nmap 10.0.0.1-50
```

Yeh range wale IP addresses (1 se 50 tak) ek saath scan kar dega.

4. Port Scanning Ke Types

4.1 TCP Connect Scan

```
nmap -sT <target>
```

Yeh sabse basic scan hai. Isme Nmap poora TCP connection banata hai target ke saath (jaise normal browser karta hai). Isme root/admin permission ki zaroorat nahi hoti, lekin yeh thoda slow aur zyada 'noisy' hota hai (logs mein trace chhod deta hai).

4.2 SYN Scan (Stealth Scan)

```
sudo nmap -sS <target>
```

Isme Nmap connection ko poora complete nahi karta - sirf 'shuruaat' karta hai aur beech mein chhod deta hai. Isse scan fast hota hai aur kam trace chhodta hai. Isko chalane ke liye root/sudo permission chahiye hoti hai.

4.3 UDP Scan

```
sudo nmap -sU <target>
```

Kuch services TCP ki jagah UDP protocol use karti hain (jaise DNS, DHCP). Yeh scan specifically UDP ports check karta hai. Yeh TCP scan se kaafi slow hota hai.

4.4 Specific ports scan karna

```
nmap -p 22,80,443 <target>
```

Comma se separate karke sirf specific ports (yahan 22, 80, 443) hi check kiye jayenge.

```
nmap -p 1-100 <target>
```

Yeh port 1 se 100 tak range scan karega.

```
nmap -p- <target>
```

Yeh sabse important flag hai - iska matlab hai saare 65535 ports check karo, default wale sirf 1000 ports nahi. Isme time zyada lagta hai lekin koi bhi hidden/unusual port miss nahi hota.

4.5 Fast scan (kam common ports)

```
nmap -F <target>
```

Yeh sirf top 100 sabse common ports check karta hai - jaldi result chahiye tab useful hai.

5. Host Discovery Commands

5.1 Sirf yeh dekhna ki device online hai ya nahi

```
nmap -sn <target/24>
```

Yeh port scan bilkul nahi karta - sirf ping jaisa check karta hai ki network mein kaunse devices 'alive' (online) hain. Poore network ke devices jaldi dhundne ke liye best hai.

5.2 Ping skip karke seedha scan karna

```
nmap -Pn <target>
```

Kabhi kabhi firewall ping requests ko block kar deta hai, jisse Nmap sochta hai device 'down' hai. -Pn lagane se Nmap ping check skip karke seedha port scan try karta hai.

6. Scan Timing Aur Speed

Nmap mein 6 timing templates hote hain, T0 (sabse slow) se T5 (sabse fast) tak:

Flag	Naam	Kab Use Karein
-T0	Paranoid	Bahut zyada stealthy, IDS bypass karne ke liye (bahut slow)
-T1	Sneaky	Kaafi slow, kam detectable
-T2	Polite	Network bandwidth kam use karta hai
-T3	Normal	Default speed
-T4	Aggressive	Fast scan, stable network par recommended
-T5	Insane	Sabse fast, lekin results kabhi kabhi miss ho sakte hain

```
nmap -T4 <target>
```

Example: yeh command normal se fast speed par scan karega, jo zyadatar cases mein safe aur quick hota hai.

7. OS Aur Version Detection

7.1 Service version detect karna

```
nmap -sV <target>
```

Yeh sirf yeh nahi batata ki port khula hai - balki yeh bhi batata hai ki us port par exact kaunsa software aur uska version chal raha hai (jaise Apache 2.4.41, ya OpenSSH 8.2).

7.2 Operating System detect karna

```
sudo nmap -O <target>
```

Yeh target device ka operating system guess karne ki koshish karta hai (Windows, Linux, MacOS, etc.) network packets ke response pattern dekh kar. Yeh 100% accurate nahi hota, sirf ek estimate hai.

7.3 Aggressive scan (sab kuch ek saath)

```
sudo nmap -A <target>
```

-A flag ek saath OS detection, version detection, script scanning, aur traceroute sab enable kar deta hai. Detailed report milta hai, lekin scan time zyada lagta hai.

8. Nmap Scripting Engine (NSE)

Nmap ke andar saikdon ready-made scripts hote hain jo automatically vulnerabilities, misconfigurations, aur extra information check kar sakte hain.

8.1 Default safe scripts chalana

```
nmap -sC <target>
```

Yeh default category ke scripts chalata hai - safe aur commonly useful information nikalne ke liye.

8.2 Specific script use karna

```
nmap --script=<script-name> <target>
```

Example ke tor par agar http-title naam ka script use karna ho, toh ki jagah http-title likh sakte hain - isse target ke web page ka title pata chal jayega.

8.3 Vulnerability scan scripts

```
nmap --script vuln <target>
```

Yeh 'vuln' category ke saare scripts ek saath chala deta hai jo known vulnerabilities check karte hain target par.

9. Output Formats Save Karna

Scan result ko file mein save karna kaafi useful hota hai baad mein analyze karne ke liye.

```
nmap -oN result.txt <target>
```

Normal readable text format mein result save karta hai (jaise screen par dikhta hai, waisa hi file mein).

```
nmap -oX result.xml <target>
```

XML format mein save karta hai, jo doosre tools ke saath import karne ke liye useful hai.

```
nmap -oG result.gnmap <target>
```

Grepable format - isse 'grep' command se easily specific information search kar sakte hain.

```
nmap -oA result <target>
```

Ek saath teeno formats (Normal, XML, Grepable) save kar deta hai, 'result' naam ke prefix ke saath.

10. Firewall/IDS Evade Karne Wale Options

Yeh section sirf conceptual samajh ke liye hai - authorized penetration testing mein hi use hota hai, kabhi bhi bina permission kisi aur network par try na karein.

```
nmap -f <target>
```

Packets ko chhote fragments mein tod deta hai, jisse kuch purane firewalls detect nahi kar pate.

```
nmap -D RND:5 <target>
```

Decoy scan - yeh asli scan ke saath 5 fake (random) source addresses bhi bhej deta hai, taaki target ke logs mein confusion ho jaye ki asli scanner kaun tha.

```
nmap --data-length 25 <target>
```

Packets mein extra random data add kar deta hai taaki woh normal traffic jaisa dikhe.

11. Useful Combinations Aur Real Examples

Example 1: Poora detailed scan

```
sudo nmap -sS -sV -O -p- -T4 <target>
```

Yeh ek complete scan hai: SYN scan (-sS), service version (-sV), OS detection (-O), saare ports (-p-), aur fast speed (-T4) - sab ek saath.

Example 2: Poore network mein online devices dhundna

```
nmap -sn <network>/24
```

Sirf yeh dekhne ke liye ki poore local network mein kaunse devices abhi online hain.

Example 3: Web server check karna

```
nmap -p 80,443 -sV --script=http-title <target>
```

Yeh sirf web server wale ports (80, 443) check karega, unki version detect karega, aur website ka title bhi nikal ke dega.

Example 4: Quiet aur fast scan

```
nmap -T4 -F <target>
```

Jaldi result chahiye ho toh yeh combination fast speed aur kam common ports check karta hai.

12. Quick Reference Cheat Sheet

Command	Kaam
nmap <target>	Basic scan
nmap -sn <target>	Sirf host discovery
nmap -p- <target>	Saare 65535 ports scan
nmap -sV <target>	Service version detect
nmap -O <target>	OS detect
nmap -A <target>	Sab kuch ek saath (aggressive)
nmap -T4 <target>	Fast speed scan
nmap -Pn <target>	Ping skip karke scan
nmap -sC <target>	Default scripts chalana
nmap -oN file.txt <target>	Result file mein save

13. Ethical Aur Legal Baatein

Nmap ek bahut powerful tool hai, aur isi wajah se ise responsibly use karna bahut zaroori hai. Kuch important points yaad rakhein:

- Sirf apne khud ke devices/network par scan karein.
- Agar kisi doosre network par test karna hai, toh pehle likhit (written) permission lein.
- Bahut saare countries mein bina authorization ke kisi network ko scan karna bhi cyber law ke under offense ban sakta hai, chahe koi damage na ho.
- Legal aur safe practice ke liye TryHackMe aur HackTheBox jaise platforms use karein, jahan authorized labs available hain seekhne ke liye.
- Yeh guide sirf educational purpose ke liye banaya gaya hai, taaki Nmap ke commands aur concepts achhe se samajh aa sakein.